

Turning Words, Sentences & Documents into Vectors

RAG

Vector DB

Semantic Search

LangChain

LLMs

What Are Embeddings?

An embedding is a dense numerical vector that represents the meaning of text (or images, audio, code) in a high-dimensional space. Texts with similar meanings end up close together — this is what makes semantic search possible. A typical embedding model outputs a vector of 768–3072 floating-point numbers.

Where Do Embeddings Fit?

- **Core AI / NLP** — Embeddings are a foundational concept used everywhere in modern NLP.
- **RAG (Retrieval-Augmented Generation)** — THE central component. Documents are embedded at index time; queries are embedded at search time; nearest vectors are retrieved and fed to the LLM.
- **LangChain** — Has a full Embeddings abstraction (`OpenAIEmbeddings`, `HuggingFaceEmbeddings`, etc.) used in `VectorstoreRetriever`.
- **Vector Databases** — Chroma, Pinecone, Weaviate, Qdrant all store and search embeddings.

How Embeddings Are Created

Modern embedding models are transformer encoders trained with contrastive objectives (e.g., Sentence-BERT, OpenAI text-embedding-3). The [CLS] token output or mean-pooled layer is used as the embedding vector.

LangChain Example:

```
from langchain_openai import OpenAIEmbeddings embedder =  
OpenAIEmbeddings(model='text-embedding-3-small') vec = embedder.embed_query('What is  
attention?') # → list[float]
```

Similarity Metrics

Metric	Formula	Best For
Cosine Similarity	$\cos(\theta) = \frac{A \cdot B}{(A B)}$	Most NLP tasks; length-invariant
Dot Product	$A \cdot B$	When vectors are normalized
Euclidean Distance	$\sqrt{\sum (A_i - B_i)^2}$	Low-dim spaces, less common in NLP

RAG Pipeline — Embeddings Role

Embeddings appear at two critical points in a RAG system:

- **Indexing phase** — Each document chunk is embedded and stored in a vector DB with its text and metadata.
- **Query phase** — The user query is embedded using the same model; nearest-neighbor search (ANN) finds the top-k chunks.
- **Reranking** — A cross-encoder can rerank retrieved chunks for better precision before passing to the LLM.

Key Rule

The query and document embeddings **MUST** come from the same model. Mixing models breaks similarity scores entirely.

Popular Embedding Models (2024–2025)

- **text-embedding-3-small / large** — OpenAI, fast and high quality.
- **all-MiniLM-L6-v2** — HuggingFace, free, great for local RAG.
- **bge-m3** — BAAI, multilingual, best open-source option.
- **voyage-3** — Anthropic's embedding partner, strong on retrieval benchmarks.